



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
UNIT TEST
17-04-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS]

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

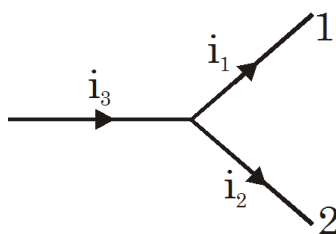
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. If a plane wavefront is incident on a concave mirror, then the emergent wavefront will be _____ of the mirror as its centre.
 (A) Spherical with centre of the curvature of the surface.
 (B) Conical with focus.
 (C) Conical with centre of the curvature of the surface.
 (D) Spherical with focus.
2. Current in the first conductor varies according to the law $i_1 = 3 \sin \omega t$ and the second $i_2 = 4 \cos \omega t$ (figure). Find i_3 .



- (A) $5 \sin\left(\omega t + \frac{37\pi}{180}\right)$ (B) $5 \sin\left(\omega t + \frac{53\pi}{180}\right)$ (C) $5 \cos\left(\omega t + \frac{37\pi}{180}\right)$ (D) $5 \cos\left(\omega t + \frac{53\pi}{180}\right)$

Space for Rough Work

3. A particle executes simple harmonic motion and it is located at $x = a$, b and c at time t_0 , $2t_0$ and $3t_0$ respectively. The frequency of the oscillation is (Take $x = 0$ as mean position & $x = A \sin \omega t$)

(A) $\frac{1}{2\pi t_0} \cos^{-1} \left(\frac{a+c}{2b} \right)$

(B) $\frac{1}{2\pi t_0} \cos^{-1} \left(\frac{a+2b}{3c} \right)$

(C) $\frac{1}{2\pi t_0} \cos^{-1} \left(\frac{a+b}{2c} \right)$

(D) $\frac{1}{2\pi t_0} \cos^{-1} \left(\frac{2a+3c}{b} \right)$

4. The magnetic field of a plane electromagnetic wave is given by $\vec{B} = B_0 \hat{i} \cos(kz - \omega t) + B_1 \hat{j} \cos(kz + \omega t)$, where $B_0 = 3 \times 10^{-5}$ T and $B_1 = 2 \times 10^{-6}$ T. The RMS value of the force (in N) experienced by a stationary charge $Q = 10^{-4}$ C at $z = 0$ is closest to

(A) 0.6

(B) 1.2

(C) 1.8

(D) 0.3

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

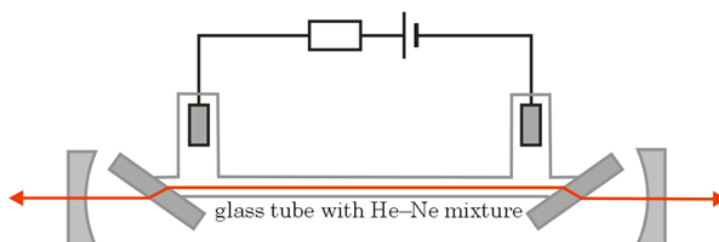
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. A Brewster window is used to minimize losses by reflection. A typical example is a helium-neon laser with a sealed glass tube as shown in Figure, where glass windows separate the laser gas mixture from ambient air. The light coming from laser is polarized. This light emanating from the glass tube strikes the two glass plates at each of the ends as shown. If the conditions are appropriate, there is no partial reflection from the glass plates. The light (shown by red colour) is then focused using convex lens shown at each of the ends. The refractive index of glass can be assumed to be 1.5 and that of both the sides of glass can be assumed to be 1.



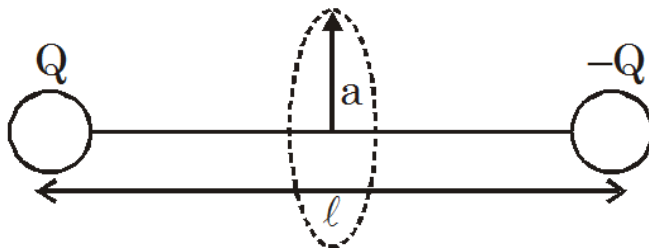
The appropriate condition(s) mentioned in the write-up is (are)

- (A) The angle made by each of the glass plate with the direction of light beam should be the same
- (B) The angle between the light beam and the glass plate surface should be $\tan^{-1}(2/3)$
- (C) The incident beam should be plane polarized with the plane of electric field same as the plane of paper
- (D) The incident beam should be plane polarized with the plane of electric field perpendicular to the plane of paper

6. A series LC circuit has a parallel plate capacitor of capacitance $40\ \mu\text{F}$ and inductance $10\ \text{mH}$. The distance between the capacitor plates is $0.1\ \text{mm}$. The resistance of circuit is negligible. The maximum charge on the capacitor is $20\ \mu\text{C}$. Choose the correct statement(s) for the average values over a long period of time:
- (A) The average force of attraction between capacitor plates is $25\ \text{mN}$.
 - (B) The average magnetic field energy is $2.5\ \mu\text{J}$
 - (C) The average energy of the electric field is 0.
 - (D) The average induced emf across the inductor is 0.
7. A series RLC circuit undergoes damped oscillation with an angular frequency of $100\ \text{rad/s}$. When the same circuit is connected to an AC source of effective voltage $200\ \text{V}$ and variable angular frequency, it shows resonance at $125\ \text{rad/s}$. The inductance is known to be $1\ \text{H}$.
- (A) The resistance of the circuit is $150\ \Omega$.
 - (B) The effective current at resonance is $17.8\ \text{mA}$
 - (C) The capacitance of the circuit is $64\ \mu\text{F}$.
 - (D) The power consumed at an angular frequency of $100\ \text{rad/s}$ is nearly $3.6\ \text{W}$
-

Space for Rough Work

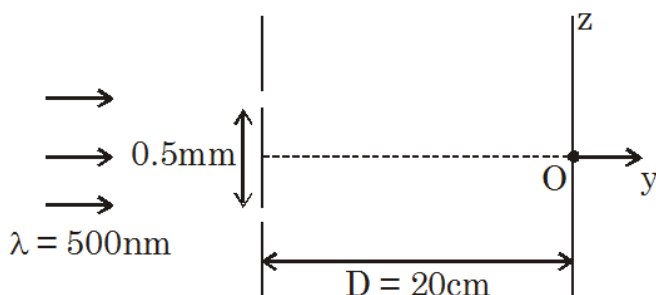
8. Consider 2 charged metallic balls of radius r connected by a thin metallic wire. The initial charge on balls is equal and opposite. Assume $r \ll \ell$ so that at any instant, the charge on the balls is spherically symmetric. Consider an amperian loop in the symmetry plane of system having radius a as shown.



- (A) The displacement current through the loop is towards left and conduction current to right.
- (B) The magnetic field at the amperian loop when current in the wire is i is $\frac{\mu_0 i}{2\pi a}$.
- (C) The magnetic field at the amperian loop when current in the wire is i is $\frac{\mu_0 i}{4\pi a} \left(\frac{\ell}{\sqrt{a^2 + \frac{\ell^2}{4}}} \right)$.
- (D) The magnetic field at the amperian loop when current in the wire is i is zero.

Space for Rough Work

9. A small charged bead of charge $+Q$, mass m is threaded on a long smooth rod oriented along y axis. An electromagnetic wave $\vec{E} = E_0 \hat{j} \sin(kx - \omega t)$ is incident on the system. Choose the correct statement (s). Neglect gravity. Initially bead was at rest.
- (A) The maximum speed of the bead is $\frac{QE_0}{m\omega}$.
- (B) The maximum speed of the bead is $\frac{2QE_0}{m\omega}$.
- (C) The average force on the bead over one time period due to magnetic field of the wave is zero.
- (D) If the bead was negatively charged, the average force on the bead over one time period due to magnetic field is in positive x direction.
10. Instead of double slits in Young's experiment, we have 2 pin holes. Assume that screen is parallel to the plane of pin holes, $\lambda = 500 \text{ nm}$, $D = 20 \text{ cm}$ & $d = 0.5 \text{ mm}$. The screen is xz plane and centre of screen is origin.

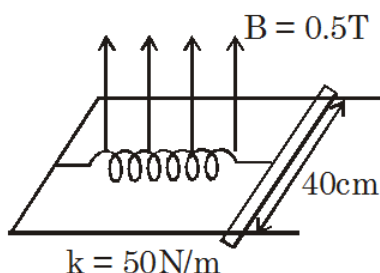


At which of the following points do we observe a maxima ?

- (A) (15 cm, 0, 0.5 mm) (B) (15 cm, 0, 0.25 mm)
- (C) (48 cm, 0, 0.52 mm) (D) (48 cm, 0, 0.26 mm)

Space for Rough Work

11. A rod of mass 400 gm is kept on a smooth resistanceless conducting loop as shown. The rod has length of 40 cm and resistance of 0.01Ω . A uniform magnetic field 0.5 T is present perpendicular to the plane of the loop. When the rod is given an initial velocity of 10 m/s to right, it executes damped oscillations. Choose the correct statement(s) :-



- (A) The angular frequency of oscillations is 10 rad/s.
- (B) The initial amplitude is 1m.
- (C) The amplitude after one oscillation is $\frac{1}{e^\pi}$ m.
- (D) The time period of oscillation decreases exponentially with time.
12. Mark the correct statement(s) :-
- (A) When a sound wave strikes a wall, the compression pulse is reflected as compression pulse.
- (B) When a sound wave strikes a wall, the compression pulse is reflected as rarefaction pulse.
- (C) When a sound wave is coming out after passing through a narrow pipe, then reflection would be there at open end.
- (D) When a sound wave is coming out after passing through a narrow pipe, then compression pulse is reflected as rarefaction pulse.

Space for Rough Work

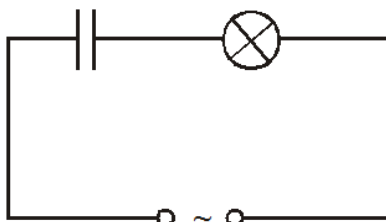
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
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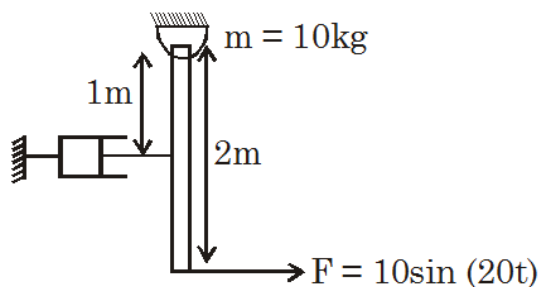
1. A particle of mass m is allowed to oscillate near the point of minima of a vertical parabolic path having the equation $x^2 = 4ay$, where x -axis is along the horizontal direction and y - axis is along the vertical direction. The angular frequency (in rad/s) of small oscillations of the particle is (take $a = 0.8$ m)
2. In a network with ac voltage of 127 V and frequency 50 Hz we connected a light bulb from a torch and a capacitor (Figure). What should be the capacitance(in μF) so that the bulb glows normally? Bulb of a torch is designed to withstand 3.5 V and current of 0.28 A.



3. A point like sound source emits sound in all direction uniformly. An observer who is at a distance of 50 m from the source detects sound of intensity 10^{-8} watt/m². If the bulk modulus of air is 1.6×10^5 N/m² and velocity of sound is 320 m/s find pressure amplitude at the position of observer. If your answer is $x \times 10^{-3}$ N/m² fill value of $[x]$ _____

Space for Rough Work

4. A laser beam of power 27 mW has a cross-sectional area of 10 mm^2 . The magnitude of the maximum electric field (in kV/m) in this electromagnetic wave is given by [Take : $\epsilon_0 \approx 9 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$ and $c = 3 \times 10^8 \text{ ms}^{-1}$]
5. A stationary source is emitting sound at a fixed frequency f_0 , which is reflected by two cars approaching the source. The difference between the frequencies of sound reflected from the cars is 1.2% of f_0 . What is the difference in the speeds of the cars (in km per hour) to the nearest integer? The cars are moving at constant speeds much smaller than the speed of sound which is 330 ms^{-1} . Round off to nearest integer.
6. A vertical rod of mass 10 kg is subjected to a sinusoidal horizontal force at its end as shown. A dashpot is attached at 1m from the hinge. The force exerted by dashpot is $F = -4v$ (in SI units) where v is velocity of piston attached to it. What is the amplitude of it's steady state oscillations (in mrad) ?



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

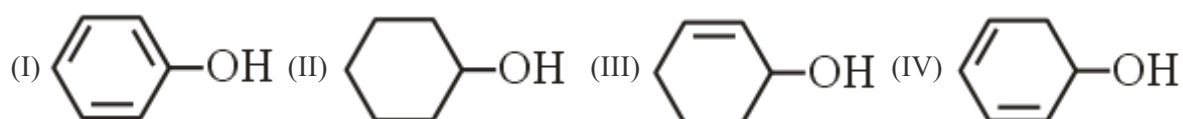
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Negative Marks : -1 In all other cases

1. Among the given compounds, the correct dehydration order is :



(A) I < II < III < IV

(B) II < III < IV < I

(C) I < III < IV < II

(D) I < II < III = IV

2. In a hypothetical reaction, $A(aq) \rightleftharpoons 2B(aq) + C(aq)$ (1st order decomposition). 'A' is optically active (dextro-rotatory) while 'B' and 'C' are optically inactive but 'B' takes part in a titration reaction (fast reaction) with H_2O_2 . Hence, the progress of reaction can be monitored by measuring rotation of plane polarised light or by measuring volume of H_2O_2 consumed in titration. In an experiment the optical rotation was found to be $\theta = 40^\circ$ at $t = 20$ min and $\theta = 10^\circ$ at $t = 50$ min. from start of the reaction. If the progress would have been monitored by titration method, volume of H_2O_2 consumed at $t = 15$ min. (from start) is 40 ml then volume of H_2O_2 consumed at $t = 60$ min will be :

(A) 60 ml

(B) 75 ml

(C) 52.5 ml

(D) 90 ml

Space for Rough Work

3. Virial equation of state can be written in terms of molar volume (V_m) or in terms of pressure as

$$Z = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \frac{D}{V_m^3} + \dots \quad (1)$$

$$\text{or } Z = 1 + B'p + C'P^2 + \dots \quad (2)$$

Find C' in terms of B and C at temperature T .

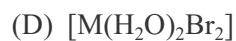
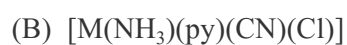
$$(A) \quad C' = \frac{C - B^2}{(RT)^2}$$

$$(B) \quad C' = \frac{C - B}{(RT)^2}$$

$$(C) \quad C' = \frac{C}{RT}$$

$$(D) \quad C' = \frac{C}{(RT)^2}$$

4. Which of the following complexes has same number of stereoisomeric structures with any geometry (tetrahedral or square planar) ?



Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

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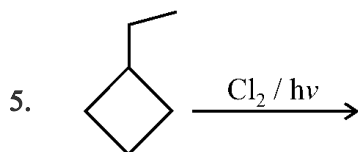
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

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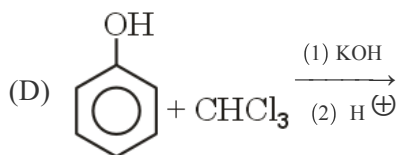
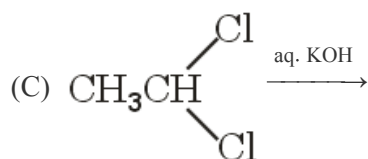
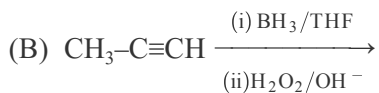
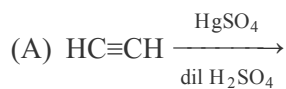
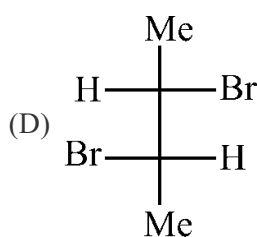
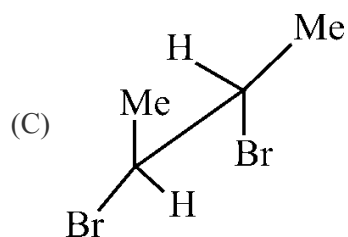
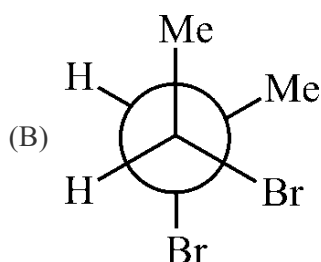
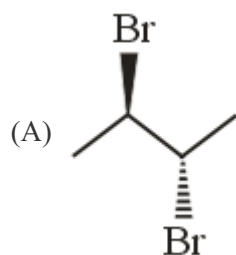
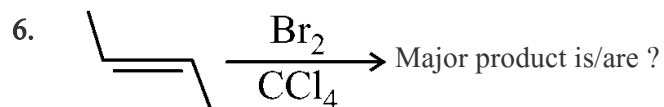
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Correct statement about monochlorinated products of above reaction :

- (A) Total 10 monochlorinated products are formed
- (B) 6 resolvable products are formed
- (C) 4 enantiomeric pair are obtained
- (D) 7 fractions are obtained after fractional distillation of product mixture

Space for Rough Work



Space for Rough Work

8. A solution contains Na_2CO_3 and NaHCO_3 . 10 ml of this solution requires 2.5 ml of $\frac{1}{5}$ M HCl solution for the end point using phenolphthalein as indicator. In another experiment 10 ml of the same original solution requires 7.5 ml of $\frac{1}{5}$ M HCl solution for the end point using methyl orange as indicator. Which of the following statement(s) is/are correct regarding the original solution.
- (A) 10 ml of original solution contain 0.053 g Na_2CO_3
- (B) 10 ml of original solution contain 0.042 g NaHCO_3 .
- (C) Concentration of Na_2CO_3 in original solution 0.05M
- (D) Concentration of NaHCO_3 in original solution 0.05M
9. In an FCC unit cell a cube is formed by joining the centers of all the tetrahedral voids to generates a new cube. Then the new cube would contain voids as :
- (A) 1 full tetrahedral void
- (B) 1 full octahedral void
- (C) 8 full tetrahedral void and 1 full octahedral void
- (D) 1 full octahedral void only

Space for Rough Work

10. Which of the following statement(s) is/are **INCORRECT**?
- (A) $[\text{Pt}(\text{PPh}_3)_4]$ is diamagnetic
(B) $[\text{Pt}(\text{PPh}_3)_4]$ is dsp^2 hybridised
(C) XeO_4 is a square planar
(D) XeO_4 and $[\text{Pt}(\text{PPh}_3)_4]$ have same structure
11. Which of the following minerals do/does not contain iron ?
- (A) Rutile (B) Ilmenite (C) Cerrusite (D) Pyrolusite
12. Compound $[\text{Co}(\text{gly})_2(\text{CN})(\text{Br})]^+$ ion has :
- (A) only one optically inactive isomer (B) two linkage isomers
(C) six geometrical isomers (D) eleven stereoisomers
-

Space for Rough Work

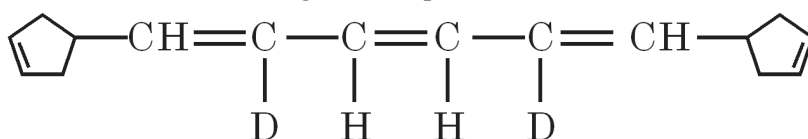
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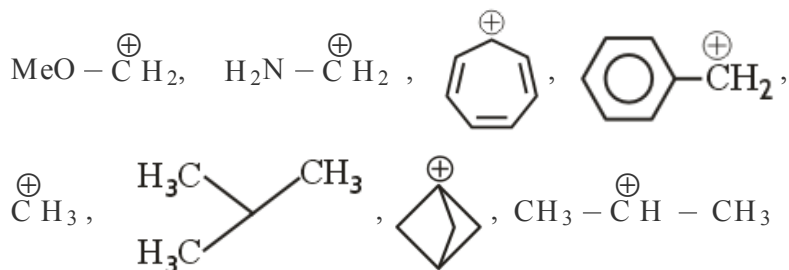
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Total number of geometrical isomers in the given compound are :-



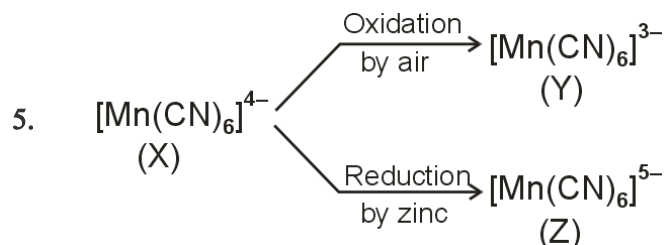
2. Identify total number of carbocation more stable than $\text{CH}_3 - \overset{\oplus}{\text{C}}\text{H}_2$?



3. At 300K, 40 ml of $\text{O}_3(\text{g})$ dissolves in 100 g of water at 1.0 atm. What mass of ozone (in gram) dissolved in 1600 g of water at a pressure of 4.0 atm at 300 K ?
 [Report your answer nearest integer value.]

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4. Calculate ΔG for the change
 $\text{H}_2\text{O}(\text{l}, 2 \text{ mole}, 1 \text{ atm}, 100^\circ\text{C}) \rightarrow \text{H}_2\text{O}(\text{g}, 2 \text{ mole}, 2 \text{ atm}, 100^\circ\text{C})$
 Assume $\text{H}_2\text{O}(\text{g})$ to behave as an ideal gas. Give your answer in cal
 [Given $\ln 2 = 0.7$, $R = 2 \text{ cal K}^{-1} \text{ min}^{-1}$].



if

the Crystal Field Stabilisation Energy of X is $= a \cdot \Delta_o$

the Crystal Field Stabilisation Energy of Y is $= b \cdot \Delta_o$

the Crystal Field Stabilisation Energy of Z is $= c \cdot \Delta_o$

(Neglect pairing energy)

then the value of $(a-b)/c$ is

6. In SbF_5 molecule, if the number of triangular planes consisting of only fluorine atoms is 'X' and the number of planes consisting of maximum number of atoms is 'Y', then find the value of (X/Y) is?

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PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. The number of integral values of k for which the equation $e^x = \frac{k}{x-3}$, has exactly two solutions is -

(A) 5	(B) 6
(C) 7	(D) 9
2. The total number of solutions of the equation $[x]^2 = x + 2\{x\}$, (where $[.]$, $\{.\}$ denotes greatest integer function and fractional part function respectively)

(A) 2	(B) 4
(C) 6	(D) 5

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3. The value of $\lim_{x \rightarrow 0} \frac{(1 + 3x + 2x^2)^{1/x} - (1 + 3x - 2x^2)^{1/x}}{x}$ is
- (A) $-e^3$ (B) $3e^2$ (C) $\frac{1}{3} e^{-2}$ (D) $4e^3$
4. Which of the following holds good for the function $f(x) = 2x - 3x^{2/3}$
- (A) the function has an extremum at $x = \left(\frac{3}{2}\right)^3$
- (B) the function has a critical point at $x = -1$
- (C) graph of $f(x)$ is concave down at every point in $(-\infty, \infty)$
- (D) graph of $f(x)$ is concave upward at every point in $\left(0, \left(\frac{3}{2}\right)^3\right)$

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SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. Let $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$, ($a > 0$) be the equation of curve then which of the following is/are correct

(A) Equation of tangent at (x_1, y_1) on the curve is $\frac{x}{x_1^{1/3}} + \frac{y}{y_1^{1/3}} = a^{2/3}$

(B) Length of portion of tangent intercepted between the coordinate axes is constant.

(C) If tangent at any point on the curve meet the coordinate axes in A and B, then locus of midpoint of AB is $x^2 + y^2 = \frac{a^2}{4}$

(D) Equation of normal at (x_1, y_1) on the curve is $xx_1^{1/3} - yy_1^{1/3} = x_1^{4/3} + y_1^{4/3}$

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6. Let a differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that for all x and y in $\mathbb{R}$ we have $2|f(x) - f(y)| < |x - y|$ and $f'(x) \geq \frac{1}{2}$. Then number of points of intersection the graph of $y = f(x)$ with
- (A) the line $y = x$ is one (B) the curve $y = -x^3$ is one
 (C) the curve $2y = |x|$ can be zero (D) the curve $y^2 = -x$ may be more than one
7. Consider $f(x) = (x - 1) \tan^{-1} x - \frac{1}{2} \ln(1 + x^2)$, $x \geq 1$. Identify which of the following statement(s) is(are) correct ?
- (A) $\lim_{x \rightarrow \infty} f'(x) = \frac{\pi}{2}$ (B) $f'(x)$ is increasing in $[1, \infty)$
 (C) $f(x + 4) - f(x) > 2\pi \forall x \in [1, \infty)$ (D) $f(x + 4) - f(x) < 2\pi \forall x \in [1, \infty)$
8. Consider the function $f(x) = \lim_{n \rightarrow \infty} \frac{(1 + \cos x)^n + 5 \log_e x}{2 + (1 + \cos x)^n}$, then
- (A) $f(x)$ is continuous at positive odd integral multiple of π
 (B) $f(x)$ is discontinuous at positive even integral multiple of π
 (C) $f(x)$ is discontinuous at positive odd integral multiple of $\frac{\pi}{2}$
 (D) $f(x)$ is continuous at positive even integral multiples of $\frac{\pi}{2}$

Space for Rough Work

9. Let $f(x)$ be a continuous function in $x \in [0, 5]$ and twice differentiable function in $x \in (0, 5)$ such that $f(4) = f(5) = 0$ then which of the following is/are correct ?
- (A) $\exists$ some ' α ' $\in (0, 5)$ such that $nf(\alpha) + \alpha f'(\alpha) = 0$, $n \in \mathbb{N}$
- (B) $\exists$ some ' α ' $\in (0, 5)$ such that $f'(\alpha) = 0$
- (C) $\exists$ distinct $\alpha_1, \alpha_2 \in (0, 5)$ such that $4f'(\alpha_1) + f'(\alpha_2) + f(0) = 0$
- (D) $\exists$ some ' α ' $\in (0, 4)$ such that $nf(\alpha) + \alpha f'(\alpha) = 0$; $n \in \mathbb{N}$
10. Let f be a twice differentiable function on $[0, 2]$ such that $f(0) = 0$, $f(1) = 2$, $f(2) = 4$, then -
- (A) $f'(\alpha) = 2$ for some $\alpha \in (0, 1)$ (B) $f'(x) = 0 \forall x \in (0, 2)$
- (C) $f'(\beta) = 2$ for some $\beta \in (1, 2)$ (D) $f'(\gamma) = 0$ for some $\gamma \in (0, 2)$
11. If $P_n = \lim_{x \rightarrow 0^+} (\log(\tan x) + \log(\tan 2x) + \log(\tan 3x) + \dots + \log(\tan nx) - \log x^n)$ and $P_{256} - P_{255} = kP_m$, (where $k, m, n \in \mathbb{N}$), then -
- (A) $k = 8$ (B) $k = 6$ (C) $m = 2$ (D) $m = 4$
12. If the curves $y = 2(x - a)^2$ and $y = e^{2x}$ touches each other, then ' a ' is less than -
- (A) -1 (B) 0 (C) 1 (D) 2

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SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Number of points where $f(x) = |(2^x - 1).(3^x - 3)x.(x - 1).(x - 2)|$ is not differentiable
2. Let $S_n = \sum_{k=0}^n \tan^{-1} \frac{49}{49k^2 + 63k + 57}$ then $\cot\left(\lim_{n \rightarrow \infty} S_n\right)$ is equal to
3. If minimum value of $y = \frac{(\sin^{-1}x)^3}{3} + \frac{\pi}{2} (\cos^{-1}x)^2$ exist at $x = \alpha$, then $\cos^{-1}\left(\frac{\cos^{-1}(-\alpha)}{\pi}\right)$ is equal to -
(Use $\pi = \frac{22}{7}$)
4. Let $f(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5}$ and $g(x) = f^{-1}(x)$, then $|g''(0)|$ is

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5. If a sequence is such that $a_1 = 1$ and $a_n = n(1 + a_{n-1}) \quad \forall n \geq 2$ and $n \in \mathbb{N}$, then $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{a_1}\right) \left(1 + \frac{1}{a_2}\right) \dots \left(1 + \frac{1}{a_n}\right)$ is _____.
6. If a cuboid, length of whose one edge is $\sqrt{2}$ units, is inscribed in a sphere of radius $\sqrt{2}$ units, then maximum possible volume of cuboid is : (in cubic unit)
-

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NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

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